

INNOMATICS®
RESEARCH LABS

INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

Hotel Reservation Analytics using MySQL

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SQL Project | Data Analytics Course

ABOUT ME

PROJECT TEAM MEMBER

AKSHAT JHA

Team Leader / Member - 1

EDUCATIONAL BACKGROUND

BCA - Completed (68%)

WORK EXPERIENCE

Fresher

WHY DATA SCIENCE?

"Passionate about deriving actionable business insights from raw data."

hub ONLINE PROFILES

PLATFORM
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Project Overview

WHAT

Project Scope

A comprehensive MySQL data analytics project analyzing hotel reservations.

TOOL

MySQL Workbench

Database schema creation, table modeling, and complex analytical query development.

DATA

Relational Tables

6 interconnected tables: hotels, guests, staff, rooms, bookings, and stays.

WORK DONE

Analytical Pipeline

Database setup, exploratory data analysis, and 5 multi-dimensional business analyses.

GOAL

Strategic Objective

Transform raw reservation data into clear, actionable commercial business insights.

Business Scenario

A hotel business runs many hotels. Guests book rooms through different channels, then stay, cancel or do not show up.

Which hotels and channels earn the most?

Bookings & revenue

Who are our guests and how do they book?

Individual vs Corporate

How do stays end?

Completed, Cancelled, No-show

How are staff and rooms performing?

Workload, ratings, usage

Where are the problems?

Cancellations & no-shows

Business Objectives

1. **Booking & Revenue Analysis**
   Top hotels, channels, room types and yearly trend

2. **Guest Booking Behaviour**
   Repeat guests, top spenders, Individual vs Corporate

3. **Stay Performance**
   Stay status, hotel-wise cancellations, stay duration

4. **Staff & Room Performance**
   Stays handled, staff ratings, most used rooms

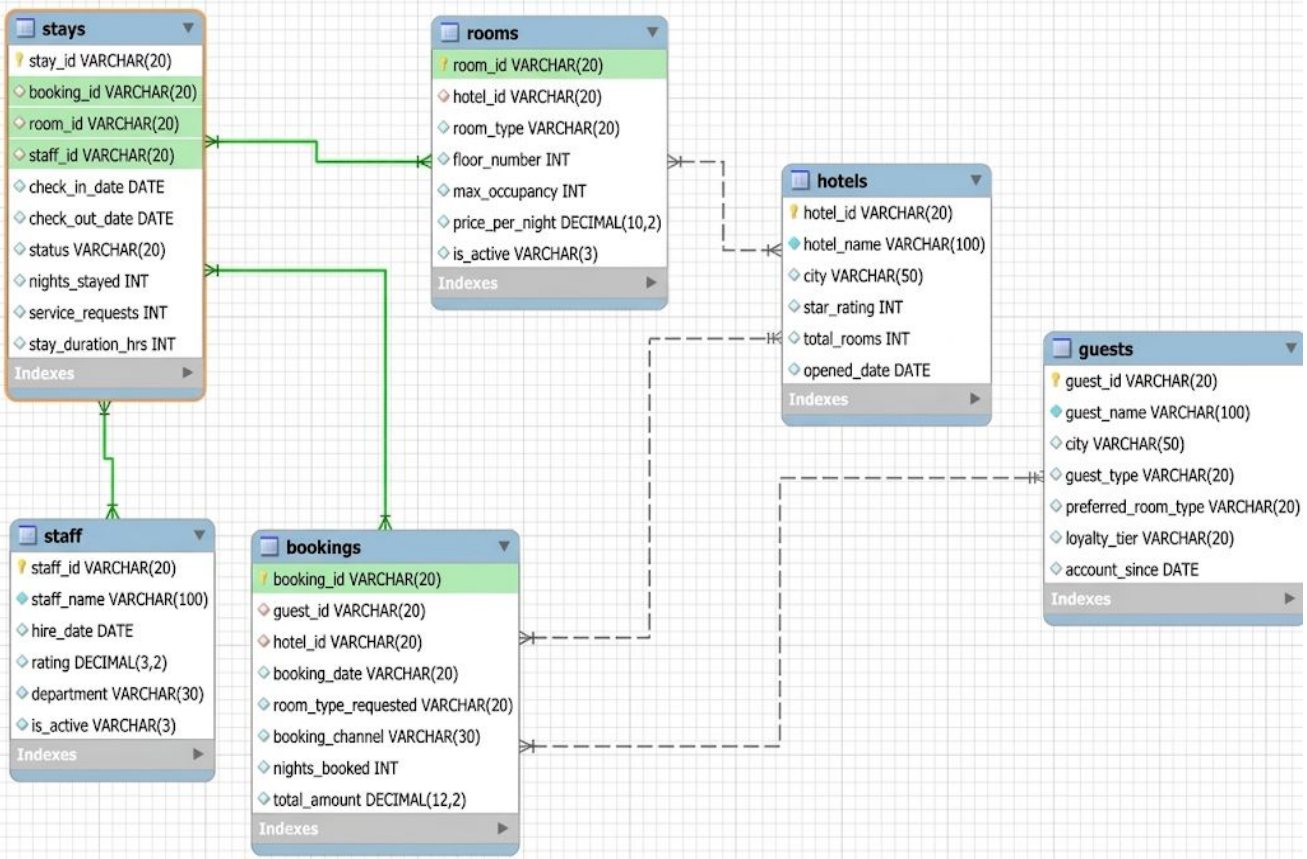
5. **Booking & Stay Problems**
   Cancellations, no-shows and service requests

Dataset & Database Overview

Table	What it stores	Key columns
hotels	Hotel details (25 rows)	hotel_id, hotel_name, city, star_rating, total_rooms
guests	Guest details (350 rows)	guest_id, guest_name, guest_type, loyalty_tier
staff	Staff details (70 rows)	staff_id, department, rating, is_active
rooms	Rooms in each hotel (600 rows)	room_id, hotel_id, room_type, price_per_night
bookings	Booking by a guest (2,800 rows)	booking_id, guest_id, hotel_id, booking_channel, total_amount
stays	Stay of a booking (3,200 rows)	stay_id, booking_id, room_id, staff_id, status, service_requests

Database name: hotel_reservation_analytics

ER Diagram & Database Structure



How the tables connect

hotels → rooms, bookings

guests → bookings

bookings → stays

rooms → stays

staff → stays

PK = Primary Key (key icon), FK = Foreign Key (red diamond). Each link is one-to-many: one parent row, many child rows.

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Basic Data Analysis (Data Exploration)

Question	Query
Total number of guests	`SELECT COUNT(*) FROM guests;`
Total number of bookings	`SELECT COUNT(*) FROM bookings;`
Total number of stays	`SELECT COUNT(*) FROM stays;`
Different room types	`SELECT DISTINCT room_type FROM rooms;`
Number of active staff	`SELECT COUNT(*) FROM staff WHERE is_active = 'yes';`
Total booking amount	`SELECT SUM(total_amount) FROM bookings;`
Average nights booked	`SELECT AVG(nights_booked) FROM bookings;`

Objective 1: Booking & Revenue Analysis

Find which hotels, channels and years bring the most bookings and money.

Q1 Which hotels get the most bookings?

SQL Query

```
select h.hotel_id,h.hotel_name,count(b.booking_id) as
total_bookings from hotels h join bookings b on h.hotel_id
= b.hotel_id group by h.hotel_id , h.hotel_name order by
total_bookings desc;
```

Output

hotel_id	hotel_name	total_bookings
H004	StayPoint Kolkata Residency	129
H006	StayPoint Mumbai Suites	128
H001	StayPoint Kochi Retreat	122
H011	StayPoint Goa Grand	122
H014	StayPoint Mumbai Grand	119
H023	StayPoint Indore Plaza	119
H024	StayPoint Pune Plaza	118
H021	StayPoint Jaipur Heights	116
H003	StayPoint Chandigarh Residency	113
H012	StayPoint Ahmedabad Suites	113
H018	StayPoint Delhi Retreat	113

What I did: Joined hotels with bookings, counted bookings per hotel and sorted high to low.

Objective 1: Booking & Revenue Analysis

Find which hotels, channels and years bring the most bookings and money.

Q2 Which booking channel earns the most revenue?

SQL Query **Output**

```
select booking_channel, sum(total_amount)
as total_revenue from bookings group by
booking_channel order by total_revenue desc;
```

	booking_channel	total_revenue
▶	Website	15298855.41
	Mobile App	15248324.38
	Travel Agent	6366745.83
	Walk-in	5205335.95
	Phone	3703917.16

What I did: Added up total_amount for each channel and sorted high to low.

Objective 1: Booking & Revenue Analysis

Find which hotels, channels and years bring the most bookings and money.

Q3 How does booking volume change by year?

SQL Query **Output**

```
select year(booking_date) as booking_year,
count(*) as total_bookings
from bookings group by year(booking_date)
 order by total_bookings desc;
```

booking_year	total_bookings
2025	1048
2024	885
2026	867

What I did: Took the year from booking_date and counted bookings in each year.

Objective 2: Guest Booking Behaviour

Understand who our guests are and how they book.

Q1 Which guests booked more than once?

SQL Query

```
select g.guest_id,g.guest_name,count(b.booking_id)
as total_booking from guests g join bookings b on
g.guest_id=b.guest_id group by g.guest_id,g.guest_name
having count(b.booking_id)>1 order by total_booking desc;
```

Output

guest_id	guest_name	total_booking
G0350	Gwendolyn Klein	64
G0265	Davis-Lewis	57
G0029	Michael Hoffman	48
G0136	David Griffith	45
G0321	Joseph Hall	44
G0316	Kelsey Parker	41
G0180	Frank Cordova	38
G0188	Danny Skinner	38

What I did: Counted bookings per guest and used HAVING to keep only repeat guests.

Objective 2: Guest Booking Behaviour

Understand who our guests are and how they book.

Q2 How do Individual and Corporate guests compare?

SQL Query **Output**

```
select g.guest_type,count(distinct g.guest_id) as total_guests,
count(b.booking_id) as total_bookings,
sum(b.total_amount) as total_booking_amount,
avg(b.total_amount) as average_booking_amount from guests g
left join bookings b on g.guest_id = b.guest_id group by g.guest_type;
```

guest_type	total_guests	total_bookings	total_booking_amount	average_booking_amount
Corporate	109	897	14428826.77	16085.648573
Individual	241	1903	31394351.96	16497.294777

What I did: Compared guests, bookings and spend by guest type. LEFT JOIN keeps guests with no bookings.

Objective 2: Guest Booking Behaviour

Understand who our guests are and how they book.

Q3 What is the average stay length by guest type?

SQL Query **Output**

```
select g.guest_type, avg(b.nights_booked)
as average_nights from guests g join
bookings b on g.guest_id = b.guest_id
group by g.guest_type;
```

	guest_type	average_nights
▶	Individual	3.3221
	Corporate	3.3088

What I did: Joined guests with bookings and averaged nights_booked for each guest type.

Objective 3: Stay Performance

See how stays end and where cancellations happen.

Q1 What share of stays falls in each status?

SQL Query

```
select status, count(*) as total_stays,
round(count(*) *100.0/(select count(*) from stays),2 )
as percentage from stays
group by status order by percentage desc;
```

Output

status	total_stays	percentage
Checked-out	2117	66.16
No-show	542	16.94
Cancelled	480	15.00
In-progress	61	1.91

What I did: Counted stays per status and used a subquery for the total to get percentages.

Objective 3: Stay Performance

See how stays end and where cancellations happen.

Q2 Which hotels have the most cancelled stays?

SQL Query **Output**

```
select h.hotel_name,count(*) as cancelled_stays
from stays s  join bookings b on s.booking_id
= b.booking_id join hotels h on b.hotel_id
= h.hotel_id where s.status ="cancelled"
group by h.hotel_name order by cancelled_stays desc;
```

hotel_name	cancelled_stays
StayPoint Mumbai Residency	45
StayPoint Mumbai Grand	30
StayPoint Indore Plaza	26
StayPoint Delhi Retreat	24
StayPoint Kolkata Residency	24
StayPoint Ahmedabad Plaza	23

What I did: Joined stays, bookings and hotels, kept Cancelled stays and counted them per hotel.

Objective 3: Stay Performance

See how stays end and where cancellations happen.

Q3 What is the average stay duration by status?

SQL Query **Output**

```
select status, avg(nights_stayed) as
average_nights_stayed,
avg(stay_duration_hrs) as
average_duration_hours from
stays group by status;
```

status	average_nights_stayed	average_duration_hours
Checked-out	3.2943	81.0340
In-progress	2.3443	57.6721
No-show	0.0000	0.0000
Cancelled	0.0000	0.0000

What I did: Averaged nights_stayed and stay_duration_hrs for each status.

Objective 4: Staff & Room Performance

Check staff workload and ratings, and which rooms are used most.

Q1 Which staff members handle the most stays?

SQL Query **Output**

```
select st.staff_id, st.staff_name, count(s.stay_id)
as total_stays_handled from staff st join stays s
on st.staff_id = s.staff_id group by st.staff_id,
st.staff_name order by total_stays_handled desc;
```

staff_id	staff_name	total_stays_handled
S044	Anthony Humphrey	59
S036	Jeffrey Chavez	57
S046	Philip Cannon	57
S027	Justin Baker	56
S049	Joshua Blair	56
S037	Sherri Baker	54

What I did: Joined staff with stays, counted stays per person and sorted high to low.

Objective 4: Staff & Room Performance

Check staff workload and ratings, and which rooms are used most.

Q2 Do staff ratings match their workload?

SQL Query

```
select st.staff_id, st.staff_name, st.department,
st.rating,count(s.stay_id) as stays_handled from staff st
left join stays s on st.staff_id = s.staff_id
group by st.staff_id,st.staff_name,st.department,
st.rating order by stays_handled desc;
```

Output

staff_id	staff_name	department	rating	stays_handled
S044	Anthony Humphrey	Housekeeping	3.94	59
S036	Jeffrey Chavez	Front Desk	4.13	57
S046	Philip Cannon	Front Desk	4.82	57
S027	Justin Baker	Concierge	3.93	56
S049	Joshua Blair	Concierge	3.15	56
S037	Sherri Baker	Housekeeping	3.76	54

What I did: Placed each staff rating next to stays handled to compare them.

Objective 4: Staff & Room Performance

Check staff workload and ratings, and which rooms are used most.

Q3 Which room types are used the most?

SQL Query

```
select r.room_type,
count(s.stay_id) as total_stays
from rooms r join stays s
on r.room_id = s.room_id
group by r.room_type order by total_stays desc;
```

Output

room_type	total_stays
Standard	1440
Deluxe	1018
Suite	464
Executive	278

What I did: Joined rooms with stays and counted stays for each room type.

Objective 5: Booking & Stay Problems

Find where cancellations and no-shows happen and what links to them.

Q1 Which booking channels have the most cancellations?

SQL Query **Output**

```
select b.booking_channel,count(*) as cancelled_stays
from bookings b join stays s on b.booking_id
= s.booking_id where s.status = "cancelled"
group by b.booking_channel
order by cancelled_stays desc;
```

booking_channel	cancelled_stays
Mobile App	164
Website	158
Travel Agent	75
Walk-in	47
Phone	36

What I did: Joined bookings with stays, kept Cancelled stays and counted them per channel.

Objective 5: Booking & Stay Problems

Find where cancellations and no-shows happen and what links to them.

Q2 Which hotels have the most problem stays?

SQL Query **Output**

```
select h.hotel_name,count(*) as problem_stays
from stays s join bookings b on s.booking_id
= b.booking_id join hotels h on b.hotel_id
= h.hotel_id where s.status in ("Cancelled","No-show")
group by h.hotel_name order by problem_stays desc;
```

hotel_name	problem_stays
StayPoint Mumbai Residency	84
StayPoint Mumbai Grand	64
StayPoint Indore Plaza	53
StayPoint Mumbai Suites	53
StayPoint Delhi Retreat	52
StayPoint Kolkata Residency	50

What I did: Counted Cancelled and No-show stays per hotel using IN.

Objective 5: Booking & Stay Problems

Find where cancellations and no-shows happen and what links to them.

Q3 Do service requests relate to stay outcomes?

SQL Query **Output**

```
select status,
avg(service_requests) as average_service_requests,
max(service_requests) as maximum_service_requests
from stays group by status;
```

status	average_service_requests	maximum_service_requests
Checked-out	1.0841	5
In-progress	0.6230	3
No-show	0.0000	0
Cancelled	0.0000	0

What I did: Took the average and maximum service_requests for each status.

Overall Key Findings

Bookings & Revenue	Top hotel: StayPoint Mumbai Grand (224 bookings). Top revenue channel: Website (15.3 million). Bookings rose from 885 (2024) to 1,048 (2025); 2026 data runs only to Aug (867).
Guest Behaviour	322 of 339 booking guests booked more than once. Individual guests spend more in total (31.4 million vs 14.4 million); average per booking is almost equal.
Stay Performance	Checked-out 66.16%, No-show 16.94%, Cancelled 15.00%, In-progress 1.91%. Most cancellations: StayPoint Mumbai Residency (45).
Staff & Rooms	Busiest staff: Anthony Humphrey (59 stays). Most used room type: Standard (1,440 stays). Staff rating has almost no link with workload.
Problems	Most cancellations: Mobile App (164); most no-shows: Website (182). Most problem stays (cancelled + no-show): Mumbai Residency (84).

Business Recommendations

- **Promote the best channels:** Website and Mobile App bring about 2/3 of bookings and revenue, so put more marketing effort there.
- **Reward repeat guests:** Offer loyalty benefits to guests who book again and again to increase lifetime value.
- **Reduce cancellations and no-shows:** About 32% of stays are cancelled or no-show. Use reminders, deposits, or flexible rebooking, starting with Mumbai Residency.
- **Balance staff workload:** Support the busiest staff and share best practices from the highest-rated staff.
- **Plan rooms smartly:** Focus on the most used room types and review rooms that are rarely occupied.

Conclusion

- **Database Architecture:** Built a robust hotel database with 6 connected relational tables in MySQL.
- **Advanced SQL Queries:** Employed JOINs, GROUP BY, HAVING, subqueries, and aggregate functions to solve key business questions.
- **Holistic Coverage:** The analysis comprehensively covered bookings, guest profiles, stays, staff workload, room performance, and operational bottlenecks.
- **Actionable Value:** Provided concrete insights to help management increase revenue, curb cancellations, and elevate guest satisfaction.
- **Professional Growth:** Significantly improved practical SQL data analytics skills and data-driven commercial acumen.

THANK YOU

Any questions?



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